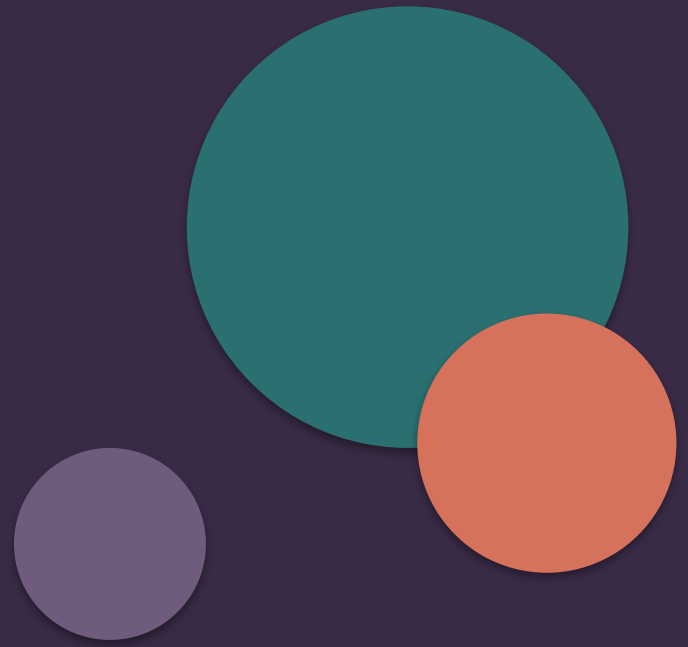
DDCA • CO5

8-bit Signed Binary Representation



How combinational circuits add, subtract and interpret signed values

K-maps • Adders • Two’s complement • ALU

Signed numbers in 8 bits

In two’s complement, the same 8-bit format supports positive and negative values.

RANGE

−128 to +127

Why this matters

The ALU works with bit patterns. Two’s complement lets the same adder hardware perform signed addition and subtraction.

Positive 13

0000 1101

Negative 13

1111 0011

Convert −x using two’s complement

Example: represent −18 using 8 bits.

- Write +18: 0001 0010
- Invert each bit: 1110 1101
- Add 1: 1110 1110
- Therefore −18 = 1110 1110

FINAL 8-BIT PATTERN

1110 1110

Half adder and full adder

Adders combine bits and produce a sum bit plus carry.

Half adder

Inputs: A, B

Sum = A XOR B

Carry = A AND B

It has no carry-in input.

Full adder

Inputs: A, B, Cin

Sum = A XOR B XOR Cin

Carry-out = AB + ACin + BCin

An 8-bit adder connects bit stages so carry can move from the least significant bit toward the most significant bit.

8-bit signed addition

Example: +12 + (−5) = +7.

+12

0000 1100

−5

1111 1011

ADD BIT PATTERNS

0000 0111 = +7

Any carry beyond bit 7 is discarded. Always check whether the mathematical answer is within −128…+127.

Subtraction using two’s complement

A − B = A + (two’s-complement of B).

- Example: 9 − 4
- 9 = 0000 1001
- 4 = 0000 0100
- Invert 4: 1111 1011; add 1 → 1111 1100 (−4)
- Add: 0000 1001 + 1111 1100 = 0000 0101
- Result: +5

Adder-subtractor

A control signal can choose B or inverted B and set the carry-in to 0 or 1, allowing one circuit to add or subtract.

Overflow in signed arithmetic

Overflow occurs when the true answer lies outside the representable range.

Overflow example

127 + 1 should be 128, but 8-bit signed range ends at +127.

BIT RESULT

```
0111 1111 + 1
= 1000 0000
```

The bit result 1000 0000 represents −128 in two’s complement, so it does not represent the true mathematical answer +128.

Where MUX, decoder and ALU fit

These combinational blocks route data, select operations and compute results.

Multiplexer

Selects one of several inputs using select lines.

Decoder

Activates an output based on an input code.

ALU

Performs operations such as ADD, SUB, AND, OR and SLT.

In an 8-bit example, the ALU processes eight-bit operands; the processor’s datapath determines where those operands and results go.

Quick practice

Try each question, then check the answers.

- 1. Write −6 in 8-bit two’s complement.
- 2. Calculate 7 + (−3).
- 3. What is the range of 8-bit signed values?
- 4. What does a MUX do?

Answers

1. 1111 1010
2. 0000 0100 = +4
3. −128 to +127
4. Selects one input

Key takeaways

Signed representation connects directly to arithmetic circuits.

- Use invert + 1 to form a negative two’s-complement number.
- Addition uses an adder; subtraction can be changed into addition.
- The 8-bit signed range is −128 to +127.
- Overflow means the correct answer is outside this range.
- MUXes route inputs, decoders select outputs, and the ALU computes.

Thank You

Combinational Logic | 8-bit Signed Binary Representation